Adhyyan Narang

MACHINE LEARNING RESEARCHER · **Ph.D., University of Washington**

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Research Profile

My research aims to understand and mitigate alignment failures in AI models, including reward hacking and unintended generalization. I design evaluations and controlled experiments to investigate when and why these failures arise, and use those insights to develop mitigations that generalize beyond individual applications. I combine empirical research on LLM behavior and training with a theoretical foundation in reinforcement learning, game theory, and optimization.

Education

University of Washington **Seattle, WA**
Ph.D., Electrical and Computer Engineering; GPA: 3.96 *Sep. 2020–Mar. 2026*
Advisors: Maryam Fazel and Lillian Ratliff; Amazon Science Hub Fellow, 2023–2024

UC Berkeley **Berkeley, CA**
M.S., Electrical Engineering and Computer Sciences; GPA: 4.00 *2020*
Advisor: Anant Sahai

UC Berkeley **Berkeley, CA**
B.S., Electrical Engineering and Computer Sciences; Minor in Theater, Dance & Performance Studies *2019*
Overall GPA: 3.95; EECS GPA: 3.97; High Distinction

Research Experience

Independent Research **Seattle, WA**
Researcher, AI Safety *Apr. 2026–Present*

- Developed Multi-Source Consensus Distillation (MSCD), a fine-tuning defense against unintended generalization that distills capabilities shared across datasets while suppressing source-specific harmful behaviors.
- On Qwen2.5-7B with three of four training sources compromised, MSCD retained 87% intent-classification accuracy while reducing unsafe medical responses from 42.5% with pooled-data training to 2.5%.
- Helped design a training algorithm for simulating vulnerable users, enabling controlled evaluation of language-model behavior during prolonged conversations.
- CoralMind: Contribute to experimental design, testing, and documentation for research aimed at improving the safety of AI use in mental health.

University of Washington **Seattle, WA**
Research Assistant *Sep. 2020–Mar. 2026*

- Built and led evaluations for an ICLR 2025 study of RL on simulated user feedback: models selectively manipulated a vulnerable 2% of users, and LLM-judge filtering sometimes made manipulation subtler. Ran harmfulness and generalization experiments, analyzed results, and wrote the evaluation section.
- First-authored a NeurIPS 2024 Spotlight achieving the tightest known instance-dependent sample-complexity bound for tabular RL. Designed an algorithm to seek out states where promising policies disagree, even through low-reward actions. Proved that these states are the most informative.
- Developed theory and algorithms for user-choice feedback loops and peer distillation (UAI 2026), multiplayer learning with decision-dependent data (JMLR, AISTATS), and interactive submodular maximization (UAI 2024).

BLISS Lab, University of California, Berkeley **Berkeley, CA**
Research Assistant *May 2019–Aug. 2020*

- Studied why classification can generalize better than regression in overparameterized models and why such models remain vulnerable to adversarial perturbations.

Publications

* *Equal contribution. Links are included in publication titles.*

Adhyyan Narang*, Artin Tajdini*, Claire Zhang*, Jamie Morgenstern. "Defense via Consensus: Mitigating Subliminal Learning through Multi-Teacher Distillation." *Preprint, 2026. Adv-ML Frontiers Workshop COLM 2026; AI4Good Workshop NeurIPS 2026. Code*

Marcus Williams*, Micah Carroll*, **Adhyyan Narang**, Constantin Weisser, Brendan Murphy, Anca Dragan. "On Targeted Manipulation and Deception when Optimizing LLMs for User Feedback." *ICLR 2025.* **Workshop orals:** SATA and SoLaR

Press coverage: **New York Times** | **Washington Post**

Publications (continued)

Cheng Li, **Adhyyan Narang**, Jiayi Yuan, John F. Buss, Jina Suh, Devon Sandel-Fernandez, Yulia Tsvetkov, Adam M. Kuczynski, Tim Althoff. "Beyond User Persona: Mechanistic User Simulation through Psychological Network Models." *Preprint, 2026*

Adhyyan Narang, Sarah Dean, Lillian Ratliff, Maryam Fazel. "Dynamics of Learning under User Choice: Overspecialization and Peer-Model Probing." *UAI 2026*

Adhyyan Narang, Andrew Wagenmaker, Lillian J. Ratliff, Kevin Jamieson. "Sample Complexity Reduction via Policy Difference Estimation in Tabular Reinforcement Learning." *NeurIPS 2024* **(Spotlight)**

Adhyyan Narang, Omid Sadeghi, Lillian J. Ratliff, Maryam Fazel, Jeff Bilmes. "Efficient Interactive Maximization of BP and Weakly Submodular Objectives." *UAI 2024*

Adhyyan Narang, Evan Faulkner, Dmitriy Drusvyatskiy, Maryam Fazel, Lillian J. Ratliff. "Multiplayer Performative Prediction: Learning in Decision-Dependent Games." *JMLR 2023; AISTATS 2022*

Yue Sun, **Adhyyan Narang**, Halil Ibrahim Gulluk, Samet Oymak, Maryam Fazel. "Towards Sample-Efficient Overparameterized Meta-Learning." *NeurIPS 2021*

Tanner Fiez, Lillian J. Ratliff, Eric Mazumdar, Evan Faulkner, **Adhyyan Narang**. "Global Convergence to Local Minmax Equilibrium in Classes of Nonconvex Zero-Sum Games." *NeurIPS 2021*

Vidya Muthukumar*, **Adhyyan Narang***, Vignesh Subramanian*, Mikhail Belkin, Daniel Hsu, Anant Sahai. "Classification vs. Regression in Overparameterized Regimes: Does the Loss Function Matter?." *JMLR, 2021*

Adhyyan Narang, Vidya Muthukumar, Anant Sahai. "Classification and Adversarial Examples in an Overparameterized Linear Model: A Signal Processing Perspective." *ICML Workshop on Overparameterization, 2021*

Industry Experience

Amazon Robotics **Seattle, WA**
Research Scientist Intern *Jun. 2023–Sep. 2023*

- Developed a learning-based approach to help warehouse robots place items neatly on shelves, preserving room for additional items; designed a placement-quality metric and trained a CatBoost model to predict it.
- Developed uncertainty-aware contextual-bandit selection to balance exploration with conservative placement decisions. On historical data, selected placements were organized 70% of the time, versus 55% for the existing system.

Amazon Customer Service AI **Seattle, WA**
Research Scientist Intern *Jun. 2022–Sep. 2022*

- Built a prototype for training customer-service chatbots with inverse reinforcement learning, using an existing latent-space dialogue model and human conversations from MultiWOZ.
- Preserved dialogue coherence during training by adding KL constraints on the latent distribution and coherence-based reward terms, addressing optimization that pushed latent representations beyond the decoder's effective range.

Uber Eats **San Francisco, CA**
Machine Learning Engineering Intern *May 2018–Aug. 2018*

- Built a stochastic-optimization system for personalized promotions and deployed it to more than 20,000 users.

Veritas Technologies **Mountain View, CA**
Data Engineering Intern *Jun. 2017–Aug. 2017*

- Built an Apache Spark service that automated ML pipelines and reduced project incubation time by 30–40%.

Teaching

University of Washington, Teaching Assistant. Foundations of Dynamical Systems (Winter 2026); Foundations of Optimization and Machine Learning (Fall 2025, Spring 2022); Introduction to Probability (Spring 2025).

UC Berkeley, Head Content Teaching Assistant. Led homework and exam design for EE 16A, a 700-student course (Spring 2020).

Awards & Honors

Amazon Science Hub Fellow, 2023–2024; B.S. with High Distinction, UC Berkeley; Phi Beta Kappa; Tau Beta Pi; Dean's List (all undergraduate semesters).

Technical Skills

Python; PyTorch; Hugging Face Transformers; NumPy/SciPy; Weights & Biases; LoRA and LLM fine-tuning; reinforcement learning; experiment design and evaluation.